

Lagged cumulative spruce budworm defoliation affects the risk of fire ignition in Ontario, Canada

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Abstract. Detailed understanding of forest disturbance interactions is needed for effective forecasting, modelling, and management. Insect outbreaks are a significant forest disturbance that alters forest structure as well as the distribution and connectivity of combustible fuels at broad spatial scales. The effect of insect outbreaks on fire activity is an important but contentious issue with significant policy consequences. The eastern spruce budworm (*Choristoneura fumiferana*) is a native defoliating insect in eastern North America whose periodic outbreaks create large patches of dead fir and spruce trees. Of particular concern to fire and forest managers is whether these patches represent an increased fire risk, if so, for how long, and how the relationship between defoliation and fire risk varies through space and time. Previous work suggests a temporary increase in flammability in budworm-killed forests, but regional and seasonal variability in these relationships has not been examined. Using an extensive database on historical lightning-caused fire ignitions and spruce budworm defoliation between 1963 and 2000, we assess the relative importance of cumulative defoliation and fire weather on the probability of ignition in Ontario, Canada. We modeled fire ignition using a generalized additive logistic regression model that accounts for temporal autocorrelation in fire weather. We compared two ecoregions in eastern Ontario (Abitibi Plains) and western Ontario (Lake of the Woods) that differ in terms of climate, geomorphology, and forest composition. We found that defoliation has the potential to both increase and decrease the probability of ignition depending on the time scale, ecoregion, and season examined. Most importantly, we found that lagged spruce budworm defoliation (8–10 yr) increases the risk of fire ignition whereas recent defoliation (1 yr) can decrease this risk. We also found that historical defoliation has a greater influence on ignition risk during the spring than during the summer fire season. Given predicted increases in forest insect activity due to global change, these results represent important information for fire management agencies that can be used to refine existing models of fire risk.

Key words: boreal forest; disturbance interactions; forest fire; forest management; general additive model; landscape legacies; logistic regression.

INTRODUCTION

The spatial legacies of large forest disturbances affect the behavior and consequences of other disturbances (Veblen et al. 1994, James et al. 2011). Landscape-scale disturbance interactions are mediated through time by the spatial patterns in forest composition and configuration that they leave behind. Dynamic and reciprocal feedback between ecological processes and these landscape legacies can be highly uncertain and have the potential to reduce ecosystem resilience (Buma and Wessman 2011), cause ecological “surprises” (Paine et al. 1998), and lead to alternate stable states (Jasinski and Payette 2005). Although our current understanding of the causes and consequences of many individual disturbances is well developed (Turner 2010), how they

constrain and facilitate one another, and how these critical relationships vary with spatial context, is less clear (Buma 2015).

Interactions between contemporary forest disturbances and persisting spatial legacies are expected to become increasingly uncertain given projected changes in disturbance regimes of northern forests (Dale et al. 2001). In particular, interactions between insect outbreaks and fire activity is of great concern (Pech 1987, Stocks 1987, Fleming et al. 2002, Lynch and Moorcroft 2008, James et al. 2011, Simard et al. 2011, Hicke et al. 2012, Watt 2014, Meigs et al. 2015). Forest fire activity is expected to increase significantly in the boreal forest in the next century due to climate change (Flannigan and Van Wagner 1991, Flannigan et al. 2009). Global change is also expected to increase insect outbreak activity in northern forests and extend their range northward and into novel habitats (Dale et al. 2001, Régnière et al. 2012). Although reliable forecasts have been developed for each of these disturbances on their own, we do not know how

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they will interact and what the expected consequences will be on forest structure and forest sustainability.

Developing robust statistical models to describe and predict the consequences of disturbance interactions is challenging due to the confounding effects of multiple stochastic processes across multiple spatial and temporal scales (Buma 2015). As a result, it is often difficult to identify clear cause-and-effect relationships. Nonetheless, useful descriptions of how fire risk is influenced by recent and historical insect defoliation are needed to inform forest management policies and to predict the consequences of disturbance on future ecosystem functioning.

The eastern boreal forests of Canada have been historically dominated by two large disturbances: stand-replacing fire (Bergeron et al. 2001) and periodic outbreaks of the spruce budworm (*Choristoneura fumiferana*; Royama 1984). Both disturbances affect ecosystem functioning and forest sustainability through their effects on forest structure (i.e., composition and configuration), nutrient dynamics (Foster et al. 1998), timber availability (Wallin et al. 1994), biodiversity (Fahrig 2003), carbon budgets (Dymond et al. 2010), as well as forest succession and subsequent disturbances (Turner 2010, James et al. 2011).

The link between spruce budworm defoliation and fire activity has been investigated for nearly a century (Graham 1923, Swaine et al. 1924, Prebble 1950, Baskerville 1975, Pech 1987, Stocks 1987, Fleming et al. 2002). Despite the availability of extensive spatial records on historical eastern spruce budworm defoliation and fire activity in eastern Canada, the precise nature of this link remains equivocal. For example, empirical studies have found no evidence of increased fire risk following defoliation in the closely related western spruce budworm (*Choristoneura occidentalis*; Lynch and Moorcroft 2008, Flower et al. 2014, Meigs et al. 2015). In contrast, studies on the eastern spruce budworm have found both an increased fire risk following defoliation (Stocks 1985, Fleming et al. 2002) and no appreciable effect (Pech 1987).

In the eastern spruce budworm system, there are several anecdotal examples of large, intense fires occurring after defoliation. These events include the Miramichi fire in New Brunswick of 1825 (Flieder 1971), the Isle Royale (Minnesota) fire of 1936 (Hansen et al. 1973), and the Ontario Mississagi wildfire in 1948 (Stocks and Walker 1973). Despite these examples, we still lack a general understanding of how spruce budworm defoliation affects fire risk. Identifying such general relationships is complicated by the fact that defoliation and consequent changes in forest structure and fuel loading persist for many years, even decades, after the insect outbreak subsides (Stocks 1987). The extent to which these fuels contribute to fire risk over time will also vary with local climate (e.g., Pech 1987). In Ontario, this window of increased fire risk has been estimated to be between 3 and 9 yr (Fleming et al. 2002), but it is also known to vary spatially (e.g., by ecoregion; Fleming et al. 2002) and between spring and summer fire seasons (Stocks 1987). Developing robust statistical models of the duration of

altered fire risk following defoliation, and how it varies with spatial and environmental context (e.g., climate, landscape structure), is needed to develop improved and flexible models of insect–fire interactions in response to a changing climate and to ensure effective and sustainable forest management.

In this paper, we aim to characterize the relationship between eastern spruce budworm outbreaks and forest fire activity. More specifically, we investigate how cumulative defoliation affects the probability of lightning-caused fire ignitions in Ontario, Canada. Using an extensive spatial-temporal database of fire ignitions, remotely sensed defoliation, weather, and lightning activity within a generalized additive logistic regression modeling framework, we ask the question: does the inclusion of historical spruce budworm defoliation increase our ability to predict the probability of lightning fire ignition relative to models developed using only fire weather? We compare the relative importance of defoliation at different time lags (1–10 yr pre-ignition) in two different Ontario ecoregions (east vs. west) for both the spring and summer fire seasons. Based on previous work in this system (Stocks 1987, Fleming et al. 2002), we expect that the effect of defoliation on ignitions will differ between seasons and between regions. We also expect that cumulative defoliation between 5 and 10 yr prior will increase the probability of ignition (Stocks 1987, Fleming et al. 2002), whereas recent defoliation may decrease this probability (Stocks 1987).

METHODS

Study area

We modeled fire ignition in two ecoregions within an area under full fire management in Ontario, Canada: Lake of the Woods (Ecoregion 91) and Abitibi Plains (Ecoregion 96; ESWG 1996; Fig. 1). This region of full fire management was until recently known as the “Intensive Protection Zone” (Ward et al. 2001). These ecoregions were chosen because of their differences in climate, vegetation, and fire regime and to examine how defoliation affects fire ignition in different spatial contexts. Both ecoregions have significant histories of fire and spruce budworm activity (Fig. 2; Fleming et al. 2002). Differences between the regions in terms of disturbance history include a greater frequency of fire and slightly more defoliation in the west. Additionally, Fleming et al. (2002) identified a longer window of increased fire risk following defoliation in western relative to eastern Ontario.

The Lake of the Woods ecoregion is characterized by warm summers and relatively little precipitation (500–700 mm/yr; Crins et al. 2009). Characteristic vegetation includes early successional forest such as trembling aspen (*Populus tremuloides*), paper birch (*Betula papyrifera*), and jack pine (*Pinus banksiana*), as well as late successional forest comprised of white spruce (*Picea glauca*), black spruce (*Picea mariana*), and balsam fir (*Abies balsamea*). Within an ecoregion, the fire regime

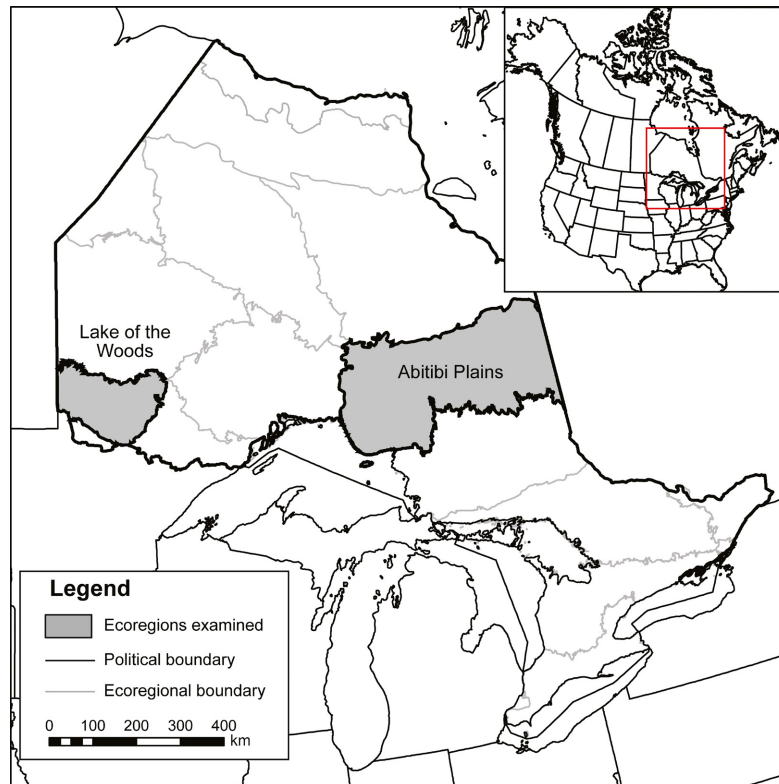


FIG. 1. Study area in the province of Ontario, Canada, showing the selected ecoregions. Upper right panel shows province in a continental context. [Colour figure can be viewed at wileyonlinelibrary.com]

varies with forest type. Upland coniferous forest fire cycles range between 50 and 187 years and fires tend to be stand-replacing. Mixed forest fire cycles are longer (63–210 yr) and fire intensity is variable (Van Sleeuwen 2006). In this region, spruce budworm outbreaks recur approximately every 31–34 yr and typically occur 6 yr after outbreaks in the Abitibi Plains (Candau et al. 1998).

The Abitibi Plains ecoregion is characterized by warm summers and cold snowy winters. The mixed forest includes white spruce, balsam fir, paper birch, and trembling aspen. This ecoregion is also warm but generally wetter than Lake of the Woods (725–900 mm/yr; Crins et al. 2009). Upland xeric sites may include jack pine or mixtures of jack pine, paper birch, and trembling aspen.

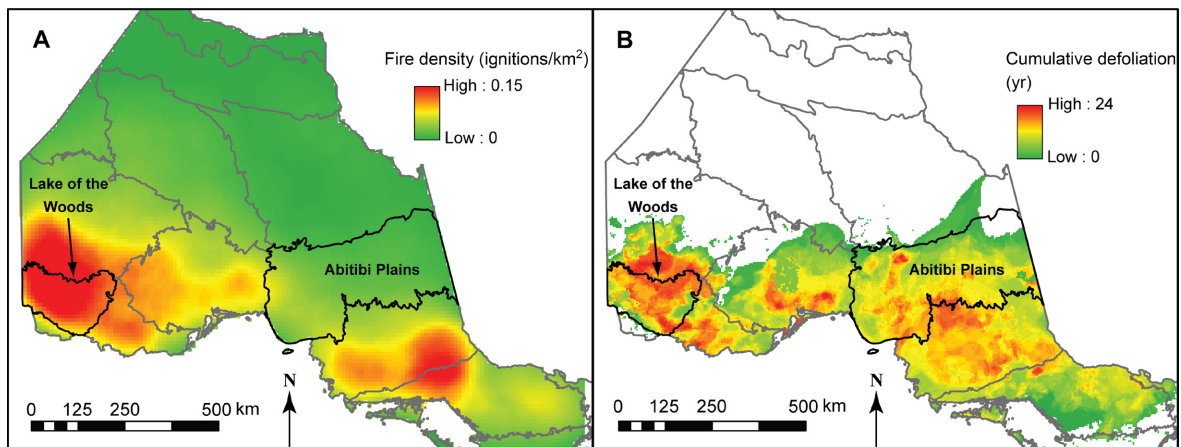


FIG. 2. Fire and eastern spruce budworm disturbance history in Ontario. (A) Historical fire activity expressed as the number of lightning-caused ignitions per square kilometer from 1963 to 2000. (B) Cumulative defoliation between 1941 and 2000. Color intensity indicates the number of years of observed moderate and severe defoliation. The two ecoregions used in this study are indicated. [Colour figure can be viewed at wileyonlinelibrary.com]

Mesic sites are characterized by black spruce and balsam fir. The fire cycle in mixed forests also ranges between 63 and 210 yr, with fires of varying intensity (Crins et al. 2009). Upland coniferous forests have a fire cycle between 30 and 187 yr and fires tend to be stand replacing. Lowland coniferous forests have much longer fire cycles, ranging between 150 and 6000 yr (Van Sleetuwen 2006). Here, spruce budworm outbreaks tend to recur every 34–37 yr (Candau et al. 1998).

Spatial data

Each ecoregion was divided into a grid of 10×10 km cells. Within each cell, we calculated multiple attributes for each year between 1963 and 2000. The resultant three-dimensional array of cells (space and time) was comprised of voxels (the three-dimensional equivalent of a pixel). Each voxel contained information on the presence or absence of fire (binary), fire weather data (six variables; Van Wagner 1987, Table 1), average lightning strike frequency, and cumulative defoliation based on aerial sketch maps from 1 to 10 yr previous. In total, the Lake of the Woods ecoregion contained 11438 voxels ($301 \text{ cells} \times 38 \text{ yr}$) and the Abitibi Plains ecoregion contained 41230 voxels ($1085 \text{ cells} \times 38 \text{ yr}$). Fire ignition data covered the years 1963–2000 and defoliation data covered the years 1941–2000. Fire weather data was available for all years and lightning strike information was available from 1992 to 2000. The average lightning strike frequency per pixel over this time period was used to control for within-ecoregion spatial variation in lightning activity.

Fire ignition.—We used information on detected lightning-caused fire ignitions of all sizes recorded by the Ontario Ministry of Natural Resources (OMNR) for the full temporal extent of the fire database (1963–2000). Each fire

event included information on the date of detection, location (latitude and longitude), and associated fire weather attributes (Van Wagner 1987). In this study, we only examined fires that were classified as lightning caused; human-caused fires were not considered. Although it is probable that some ignitions were not detected, for the purposes of this study, we assume that any errors are not systematic in such a way as to bias our results. In total, 3273 fire events were retained for analysis; 2195 in Lake of the Woods and 1078 in Abitibi Plains. Ignitions were initially represented as a point-data layer that was overlaid with the 10×10 km grid layer for each ecoregion for each year. Each cell within a single year was assigned a binary value for the presence or absence of fire. The combined binary spatial-temporal data served as the response variable in the logistic regression models described below.

Fire weather.—The Canadian Forest Fire Weather Index (FWI) System (Van Wagner 1987) uses observations of daily noon temperature, relative humidity, wind speed, and 24-h accumulated rainfall to estimate fuel moisture indicators and to calculate six fire danger rating indices (Table 1). Because the relationship between fire and weather is complex, these composite fire weather indices were developed to more effectively relate the effects of weather on fire occurrence, spread, and severity. Fire weather was summarized for two time periods within each year, a spring season (days prior to June 15) and a summer season (days after June 15; Stocks 1987). The start of the spring season, and the end of the fall season, were determined by the date of the first and last ignitions recorded, respectively. For use as model predictors, we used the 95th percentile of the distribution of each fire weather variable within each grid cell for each fire season. We used the 95th percentile rather than mean values as fires are more likely to be the product of

TABLE 1. Summary of predictor variables examined to explain spatiotemporal variation in the probability of fire ignition.

Predictor	Code	Description	Predictor variable summaries	
			Lake of the Woods	Abitibi Plains
Cumulative defoliation (1–10 yr)	Defol. <i>X</i>	cumulative percentage of cell defoliated over the last <i>X</i> years	1.49 (4.83)	1.12 (3.64)
Lightning	AVG_LTG	mean lightning strike frequency per cell (Flannigan and Van Wagner 1991)	81.19 (286.26)	34.4 (85.27)
Fine-fuel moisture code	FFMC†	forest floor litter moisture content	78.65 (26.32)	75.94 (33.02)
Drought code	DC†	moisture content of deep organic material in the forest floor and the large dead and down material	124.96 (4,610.01)	114.85 (4,958.01)
Duff moisture code	DMC†	moisture content of loosely compacted decomposing organic matter	17.22 (62.94)	13.64 (34.96)
Initial spread index	ISI†	relative fire rate of spread potential	2.66 (1.05)	2.32 (0.87)
Build-up index	BUI†	the increasing mass of fuel available as drought period progresses	23.88 (103.45)	19.25 (69.74)
Fire weather index	FWI†	the fire danger rating at its mid-afternoon peak	4.65 (8.26)	3.64 (4.72)

Notes: Summaries are based on the mean predictor values in each voxel (10×10 km resolution $\times$ 1 yr) within both ecoregions (Lake of the Woods, LotW; and Abitibi Plains, AP) from 1963 to 2000. Fire weather variables and their calculation are defined in detail in Van Wagner (1987) Values are means with SD in parentheses.

†Fire weather indices were calculated using the Canadian Forest Fire Weather Index System (Van Wagner 1987).

extreme weather events than average tendencies (Amiro et al. 2005).

Lightning.—The OMNR has operated a lightning detection system for a number of decades, and has been consistently archiving these data since 1992. In areas of intensive fire management, the spatial accuracy of this lightning detection network is thought to be approximately 1–2 km 95% of the time (Wotton and Martell 2005). We included the average number of lightning strikes per year from 1992 to 2000 for each 10 × 10 km grid cell as a predictor in our model selection procedure. Although we do not have spatial lightning strike information on an annual basis for the whole time-series of fire and outbreak data, we believe that it is reasonable to use the average lightning strike layer per grid cell from this time period to control for expected within-ecoregion spatial variation in ignitions due to lightning activity alone.

Spruce budworm defoliation.—Defoliation data were derived from aerial survey data collected by the Forest Insect and Disease Survey (FIDS) of the Canadian Forest Service that were converted to digital vector format. During survey flights, an observer classifies spruce budworm defoliation impact as light, moderate, or severe (Sippell 1983). We calculated cumulative defoliation up to 10 years prior to all ignitions in our fire data set. We limit the temporal depth of the cumulative defoliation predictor to 10 years prior to ignition based on previous work that found that the influence of spruce budworm defoliation on fuel structure and fire activity declines after 10 years (Stocks 1987, Fleming et al. 2002). For each year and for each cell we calculated the percentage of the cell that was classified as moderately or severely defoliated; light defoliation was not considered due to the uncertainty with which this category is classified and expected negligible effects on fuels. Annual defoliation percentages were then summed at different time lags (1–10 yr) resulting in 10 separate cumulative defoliation variables associated with each voxel. For example, the maximum summed percent defoliation over 10 years in

a situation where there was 100% defoliation per year would be equal to 1000%.

Analysis

Our goal was to develop a parsimonious predictive model of fire ignition using fire weather and defoliation variables. The presence and absence of ignitions within each voxel served as the response variable in a generalized additive logistic regression model framework. Prior to analysis, all predictors (Table 2) were standardized to zero mean and unit variance. Models were built separately for spring and summer fires within each of our two ecoregions (Fig. 1). A general additive model (GAM) component was included with our logistic regression using the variable “Year.” This GAM component (Zuur et al. 2009) was included to account for temporal autocorrelation in fire weather, and hence ignition probability, for each ecoregion × season combination (Fig. 3). All models were built using the mgcv package in R (Wood 2000) using the gam function with a spline-based smoother function on the variable “Year.” An information theoretic approach was then used to identify the best model that minimized the Akaike information criterion (AIC; Burnham and Anderson 2003). This criterion seeks to maximize the variance (deviance) described by the model while minimizing the number of predictors included.

Many voxels did not contain any fires and, as a result, our data are heavily zero inflated. In the Abitibi Plains, 97.41% of voxels had no ignitions; in the Lake of the Woods, 81.77% had no ignitions. Zero-inflated data in logistic regression can result in significant underestimation of rare events, such as fires (King and Zeng 2001). To reduce the bias associated with zero inflation, we retained only those cells (i.e., geographic locations) that had at least one recorded lightning-caused ignition during the time span of our data. That is, locations that never experienced an ignition throughout the timespan of the data were removed entirely from analysis. Similar approaches using subsets of absences in logistic regression have been previously described in King and Zeng (2001).

TABLE 2. Summary of final models for each ecoregion and season.

Ecoregion and season	Presences/ absences	Weather-only model	Defoliation-only model	Total deviance explained (%)	ΔAIC	AUC
Abitibi Plains						
Spring	182/5898	BUI	Defol.8	26.2	−5.71†	0.85
Summer	896/19320	DC + DMC	Defol.8 + Defol.1	10.2	−19.23	0.75
Lake of the Woods						
Spring	293/7041	BUI + DC	Defol.10 + Defol.1	22.1	−21.38	0.86
Summer	1902/9346	DC + FWI	Defol.10	9.86	−12.72	0.72

Notes: Presences/absences indicate the number of voxels used in model fitting that did and did not contain a record of fire ignition. Variables listed under weather-only and defoliation-only models indicate the variables included in these separate models prior to merging the two. ΔAIC shows the marginal improvement of the combined weather + defoliation model over the weather-only model. AUC is the mean area under the curve statistic from the ROC analysis and indicates model performance. Total deviance explained indicates model goodness-of-fit of the full combined model. Weather and defoliation variables are defined in Table 1. Weather, defoliation, and combined models were selected on the basis of a minimized AIC.

†Because ΔAIC < 10, we cannot distinguish which is the better model.

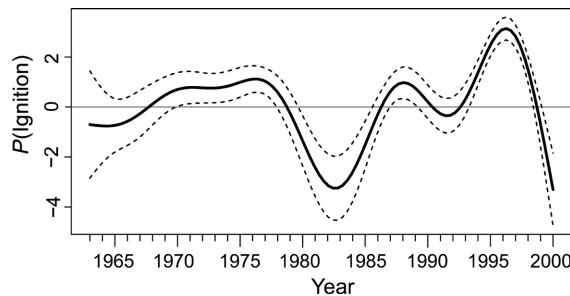


FIG. 3. General additive model (GAM)-estimated temporal variation in the probability of fire ignition ($P(\text{ignition})$) using spring fires in the Abitibi Plains ecoregion as an example. Solid line represents the average probability of fire ignition across the ecoregion as a function of time in the logit scale; dashed lines represent the 95% confidence interval ($1.96 \times \text{SE}$). Curve is the nonlinear smoother of the GAM model as fitted using regression-based splines on the variable “Year.” Equivalent curves were generated for all ecoregion $\times$ season combinations. This smoothed temporal component was included in all models to control for known temporal autocorrelation in fire weather. In this case, the overall estimated degrees of freedom (EDF) of the GAM model was high ($\text{EDF} = 8.86$; $\chi^2 = 210.2$; $P < 0.01$), which shows strong evidence against using a regular GLM to describe temporal variation in ignitions (Zuur et al. 2009). Similar significant EDF values were found for all other ecoregion $\times$ season combinations (not shown).

Model building process.—Models were built in three steps. We first constructed weather-only models for each ecoregion $\times$ season combination to predict fire ignition as a function of all fire weather variables and lightning strike frequency using AIC-based automatic forward selection. We then sought to explore the extent, if any, to which these models could be improved using data on budworm defoliation. We next constructed a set of defoliation-only models (no weather) to predict ignitions, also using AIC-based forward selection on the set of 10 cumulative defoliation variables (i.e., 10 time lags). Finally, we combined the predictors retained in these individual “best” models and reran the step-wise selection procedure to identify a best overall model when considering weather and defoliation together. Models combining weather and defoliation with a $\Delta\text{AIC} > 10$ (Burnham and Anderson 2003) relative to the weather-only model are considered as evidence that defoliation improves models of fire ignition.

Collinearity in predictors can confound interpretation and the signs of individual coefficients. We used the Variance Inflation Factor (VIF) to reduce model collinearity; variables that exhibited a $\text{VIF} > 3$ were removed from the final models. VIF was calculated using the *car* package in R (Fox and Weisberg 2010). To demonstrate the value of including defoliation in the models, we also examined the marginal effect of each cumulative defoliation variable on the AIC of the selected weather-only model.

Model performance.—Model performance was assessed using a bootstrapped receiver operating characteristic (ROC) analysis and the area under the curve (AUC) statistic (Fielding and Bell 1997). ROC analysis classifies model

predictions into one of four categories: true positive (TP), false positive (FP), true negative (TN), and false negative (FN). In this study “positives” represent ignitions and “negatives” represent the absence of ignition. These four values are combined into a 2×2 confusion matrix from which the true positive rate ($\text{TP}/[\text{TP} + \text{FN}]$) and false positive rate ($\text{FP}/[\text{FP} + \text{TN}]$) can be calculated. Because classification into these categories is sensitive to the threshold predicted value that is used to distinguish a positive from a negative, threshold independent measures of model performance such as ROC are generally preferred (Liu et al. 2005). The ROC curve is produced by plotting the true positive rate as a function of the false positive rate for all possible thresholds. An ROC curve can then be summarized using the AUC statistic that indicates the probability that a model will correctly predict event occurrence; an AUC of 1 indicates perfect discrimination between positives and negatives, whereas an AUC of 0.5 indicates discrimination no better than random chance.

Data associated with each of our four models were split into training (70%) and validation (30%) sub-sets. Due to the paucity of fire events relative to non-fire events (i.e., empty voxels), a completely random spatiotemporal sample of 70% of our sites has a high probability of selecting very few fires (i.e., presences) and biasing estimates of model performance. To overcome this challenge, data splitting was based on geographic location only. That is, we randomly selected 70% of sites (i.e., 10×10 km cells) that had experienced at least one fire historically, and fit the models using all temporal data (i.e., voxels of presences and absences) associated with those sites. We then assessed how well the models that were fit predicted the spatiotemporal pattern of fire ignitions in the remaining 30% of sites using ROC. In this way we avoided the situation where a testing or a validation subset could be overly dominated by absences. This procedure was repeated 1000 times with different validation and testing data sets. ROC analysis was undertaken using the ROCR package in R (Sing et al. 2005).

RESULTS

Final models included fire weather variables and cumulative defoliation variables at 1, 8, and 10 yr prior to ignitions for spring and summer fire seasons in both ecoregions (Table 2, Fig. 4). In all models, adding selected cumulative defoliation variables improved model fit (i.e., reduced AIC; Fig. 5). Although the marginal contribution of defoliation at a lag of 1 year did not initially improve model fit, it did substantially reduce AIC when used in combination with defoliation at lags of 8 and 10 years (i.e., stars in Fig. 5). The magnitude of coefficients associated with defoliation was generally lower than those associated with fire weather. Deviance explained by each model was generally lower in the summer than in the spring (Table 2). We did not find any effect of lightning strike frequency, nor did we find evidence for interactions between fire weather and

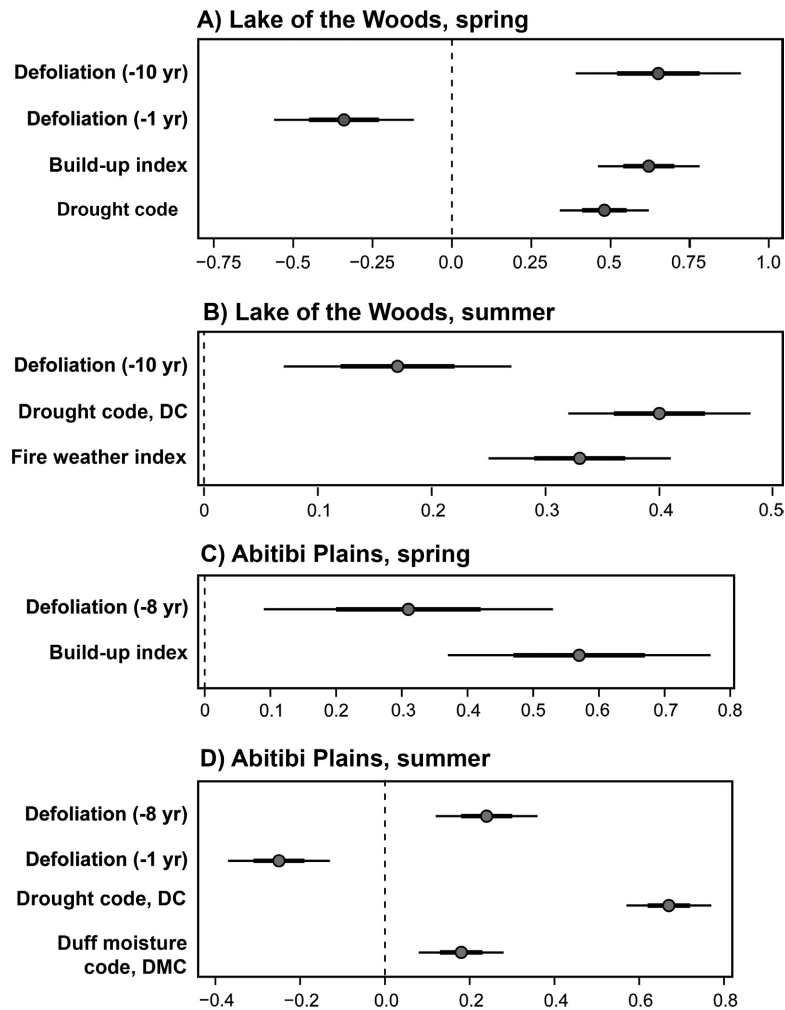


FIG. 4. Coefficient plot for predictors of each of our four models: (A–D). Thick and thin error bars represent 1 and 2 SD, respectively. All included predictors are statistically significant at $\alpha = 0.05$. Note that although defoliation is included in the model for Abitibi Plains in spring (C), we were not able to distinguish the weather-only model from the weather + defoliation model on the basis of ΔAIC (the change in the Akaike information criterion due to the model).

cumulative defoliation (*not shown*). We summarize each ecoregion $\times$ season model below.

Abitibi Plains, spring

Our spring model for the Abitibi Plains ecoregion was built using 182 ignitions (i.e., voxels in which an ignition was present) and 5898 non-fire events (i.e., voxels in which ignitions were absent). Our final best model included cumulative defoliation at 8 yr (+) and build-up index (+), both of which had a positive influence on the probability of ignition (Fig. 4). Build-up index had a greater influence on ignition probability than defoliation (Fig. 4). Including cumulative defoliation in the model decreased the AIC of the weather-only model by 5.71 units. Because this change in AIC was small (i.e., $\Delta AIC < 10$) the two models must be considered as equivalent and we cannot conclude that defoliation affects ignitions in this case (Table 2).

Abitibi Plains, summer

Our summer model for the Abitibi Plains ecoregion was built using 896 fires and 19320 non-fire events. Our final best model included cumulative defoliation at 1 (–) and 8 yr (+) as well as drought code (+) and duff moisture code (+). Here, drought code has the greatest influence on ignitions, followed by recent (1 yr) defoliation, old defoliation (8 yr), and duff moisture code (Fig. 4). Including cumulative defoliation reduced the AIC of the weather-only model by 19.23. Defoliation at 8 yr increased the probability of fire (positive coefficient) whereas defoliation at 1 yr decreased the probability of fire (negative coefficient; Table 2, Fig. 4).

Lake of the Woods, spring

Our spring model for the Lake of the Woods ecoregion was built using 293 fire and 7041 non-fire events. Retained predictors for this model included cumulative defoliation

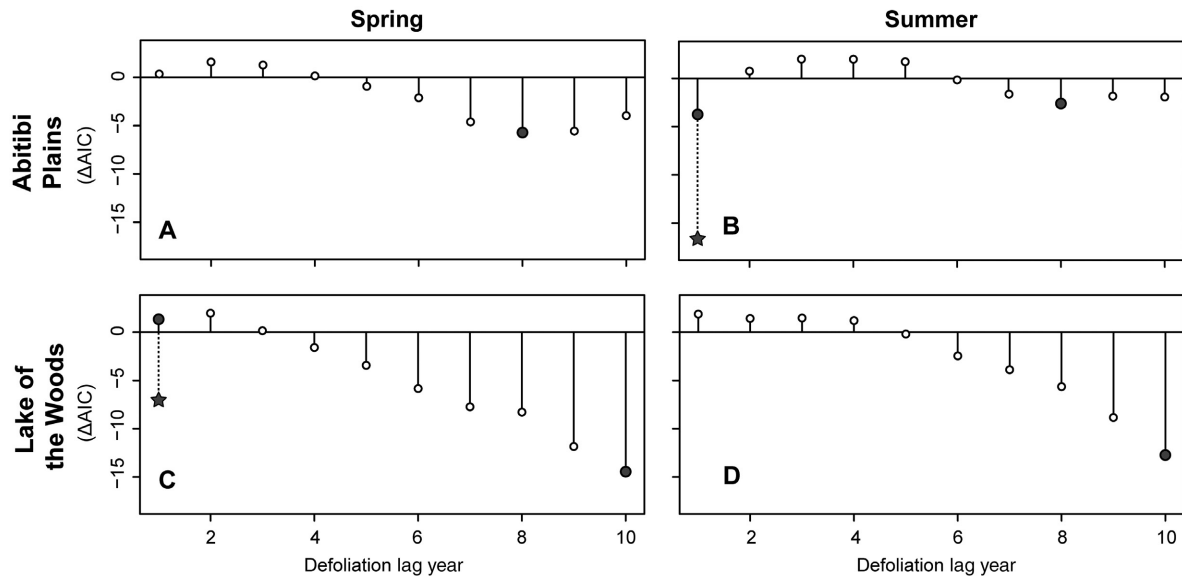


FIG. 5. Change in AIC of weather-only models following the addition of individual cumulative defoliation variables for each ecoregion $\times$ season combination: (A) Abitibi Plains $\times$ Spring; (B) Abitibi Plains $\times$ Summer; (C) Lake of the Woods $\times$ Spring; (D) Lake of the Woods $\times$ Summer. Circles indicate the marginal change in AIC when adding a single defoliation variable to weather-only models. Solid circles indicate the defoliation variables that were retained in final models. Stars indicate the further marginal reduction in AIC when adding a second defoliation variable to the weather and single defoliation variable model.

at 1 (–) and 10 yr (+), build-up index (+), and drought code (+). Here, defoliation at 10 yr had the greatest influence on ignitions, followed by build-up index, drought code, and recent defoliation (Fig. 4). Cumulative defoliation at 10 yr prior to ignition increased the probability of ignition whereas recent defoliation decreased this probability (Fig. 4). Addition of cumulative defoliation to the weather-only model decreased AIC by 21.38, making it the greatest improvement of all models (Table 2). This suggests that defoliation has a large influence on fire occurrence in this ecoregion in the spring.

Lake of the Woods, summer

The summer model for Lake of the Woods used 1902 fire and 9346 non-fire events. Retained predictors for the summer model included defoliation at 10 yr (+), drought code (+), and fire weather index (+) (Table 2, Fig. 4). Here, drought code had the greatest influence on ignitions, followed by fire weather index, and defoliation at 10 yr (Fig. 4). Including defoliation in the model decreased AIC by 12.72. In this region during the summer season, cumulative defoliation influences fire risk, but fire weather predictors have a larger effect as evidenced by their larger coefficients. (Table 2, Fig. 4).

Model performance

Models performed average to well according to ROC analysis and the AUC statistic (Fig. 6). AUC values for each season $\times$ ecoregion combination were Abitibi Plains $\times$ spring, 0.84; Abitibi Plains $\times$ summer, 0.75; Lake

of the Woods $\times$ spring, 0.85; Lake of the Woods $\times$ Summer, 0.72. In general, models for spring ignitions performed better than for summer ignitions. Model performance did not differ between ecoregions (Fig. 6).

DISCUSSION

Forest disturbances are often influenced by the legacies of previous events (Wallin et al. 1994, James et al. 2007, Turner 2010). As hypothesized (i.e., Stocks 1987, Fleming et al. 2002), we found that the spatial legacies of spruce budworm outbreaks can both increase and decrease the probability of ignition. In all cases, including information on cumulative defoliation improved models relative to a simple weather-only model (Table 2, Fig. 5). Recent defoliation was found to decrease ignition probability whereas cumulative defoliation over a longer time period increases the risk of ignition. This means that although weather has an important role in determining where and when ignitions occur, forest structure created as a result of defoliation also matters.

The specific effect of defoliation on ignition varies between ecoregions and fire seasons. In general, cumulative defoliation has a greater positive influence on the probability of ignition in dry, western regions relative to wetter, eastern regions (Table 2). In the west, we also found that cumulative defoliation has a greater positive influence on spring than summer ignitions. Although defoliation was relevant and significant in both ecoregions during the summer fire season, we were not able to conclusively identify an effect of defoliation in the Abitibi Plains ecoregion in the spring.

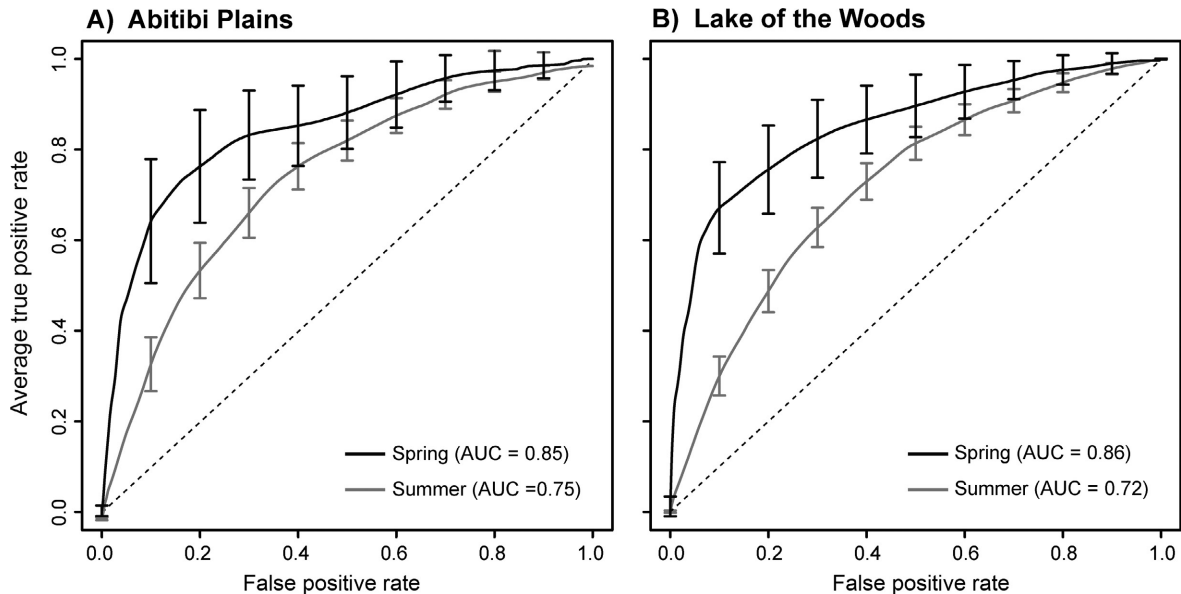


FIG. 6. Receiver operating characteristic (ROC) curves illustrating the performance of our final models in (A) Abitibi Plains and (B) Lake of the Woods. Final model performance is summarized using the area under the curve statistic (AUC) where a value of 1 indicates perfect discrimination between presences (ignitions) and absences, and 0.5 (diagonal dashed line) indicates discrimination no better than random chance.

It is thought that eastern spruce budworm defoliation creates a “window of opportunity” for fire after defoliation driven by temporally dynamic fuel structures (Stocks 1985, 1987, Fleming et al. 2002). For a period after defoliation, the breakage of dead tree tops and wind throw gradually build up an accumulation of “ladder fuels” in a stand (Stocks 1985, 1987, Watt 2014). This buildup of elevated dead fuel from canopy tops can facilitate ignitions and permit the flames from small surface fires to extend higher into the canopy (Stocks 1985, 1987, Fleming et al. 2002). This “window of opportunity” was found to persist longer in western Ontario than in the east (Fleming et al. 2002). Defoliation also alters fire risk by increasing the accumulation of needles and other dry surface fuels on the forest floor and promoting the occurrence of surface fires (Stocks 1987). The inclusion of cumulative defoliation at 8–10 yr before ignition in all models strongly supports this conceptual model of how fuel structure develops following defoliation and increases the risk of ignition. The steadily increasing ΔAIC with an increase in time lag (Fig. 5) in the Lake of the Woods ecoregion suggests that “windows of opportunity” greater than 10 yr could also be considered. While this is possible, we restrict our analysis to 10 yr given (1) previous work on the temporal dimensions of fire–eastern spruce budworm interactions (Stocks 1987, Fleming et al. 2002); (2) what is empirically known about the temporal development of forest fuel structures following defoliation (Stocks 1987); and (3) the highly correlated nature of the cumulative defoliation variables that reduces their statistically reliability at longer lags.

We found that cumulative defoliation is more important to spring than summer ignition risk, based on

the greater deviance explained in the spring models that include defoliation (Table 2). This difference is likely due to seasonal differences in the composition and moisture of the fuel complex near the surface. Fire risk is generally thought to be higher in the spring prior to the development of surface vegetation. The development of such surface vegetation and canopy leaf-out in summer can shade surface fuels and increase moisture levels. Greater humidity in the summer fire season then acts as an energy sink that reduces the combustibility of the surface and near surface layers and therefore reduces the potential for connectivity between surface and crown fuels (Stocks 1987). Thus, the increased flammability of dead fir and spruce has a greater chance to affect ignitions in the dryer spring season.

The important negative influence of defoliation at a 1-yr lag in the spring, Lake of the Woods model and the summer, Abitibi Plains model (Fig. 4) is consistent with the idea that very recent defoliation can reduce the risk of ignition through an opening of the canopy that increases site-level humidity due to proliferation of understory vegetation (Stocks 1987). We speculate that, in the years following intense defoliation and tree mortality, the accumulation of surface and elevated dead fuels eventually reach a point where fuel accumulation offsets this initial dampening effect of understory vegetation (Stocks 1987). Our results suggest that the hypothesized ignition-reducing effect of site-level green-up is most relevant immediately following a cell being classified as defoliated (i.e., ~1 yr). It is important to bear in mind that the defoliation data used here are derived from remotely sensed imagery that represent observed tree discoloration and

mortality as a result of several years of continuous defoliation. Our results suggest that, after 8–10 yr of such observable defoliation, ignition risk starts to increase due to fuel accumulation, as originally hypothesized by Stocks (1987).

That recent defoliation affects fire ignition in different seasons in the different ecoregions is somewhat perplexing but may reflect ecoregional differences in climate and weather. Low precipitation and warm temperatures in the west result in longer periods of water deficit than in the east and drier climates are more conducive to fire ignition. We suggest that the Lake of the Woods region in the west, where overall conditions are indeed drier and fire more common, has the potential to be more affected by the accumulation of fuels after defoliation than more mesic eastern regions. In other words, although the window of opportunity created by the effects of defoliation on fuels may be similar between ecoregions, there is a greater opportunity for fire to realize that opportunity in the west. It is in these dryer regions that the greatest management and policy benefits will be derived from including information on historical defoliation in models of fire risk.

Fire–insect interactions in other systems

Insect–fire interactions are not unique to the spruce budworm system. Recently, a great deal of attention has been directed toward the effects of the mountain pine beetle (MPB; *Dendroctonus ponderosae*) on fire in western North America (Simard et al. 2011, Hicke et al. 2012, Meigs et al. 2015). Some studies have concluded that disturbance by mountain pine beetle does not increase the probability of fire or area burned (Simard et al. 2011, Harvey et al. 2014, Hart et al. 2015) but may affect fire behavior under particular weather conditions (Simard et al. 2011). Studies of western spruce budworm (WSBW; *Choristoneura occidentalis*) have reported a decrease in fire risk due to insect damage across multiple ecoregions and time lags (Lynch and Moorcroft 2008, Meigs et al. 2015, 2016). Western spruce budworm defoliation has also been shown to have no influence on burn severity, possibly due to reductions in live vegetation following defoliation that reduce the amount and connectivity of fuels on the landscape (Meigs et al. 2016). These results suggest that, for these systems (i.e., MPB and WSBW), insect-caused changes in forest fuels are less important than other factors (e.g., climate and weather) to fire dynamics.

In contrast, our findings suggest that insect–fire interactions can both increase and decrease the risk of ignition in the eastern spruce budworm system and should be considered in developing future fire and forest management policies. We found that the effects of defoliation on ignition risk are more pronounced in dryer areas with a shorter fire cycle and are sensitive to both seasonality and environmental context (e.g., ecoregion). Our results clearly indicate that the spruce budworm activity

increases the risk of ignition after 8–10 yr of defoliation and that very recent defoliation can reduce the risk of ignition. Our spatially and temporally contingent results contrast those from other systems and highlight the importance of considering multiple spatial and temporal scales (i.e., time lags) when investigating disturbance interactions and developing forest management strategies.

Differences among taxa in their effects on forest fire may be due to the different way in which they convert living forest biomass into fuels. The eastern spruce budworm generally causes greater damage and higher percentage mortality than the western spruce budworm and is thought to create more abundant and more complex fuel structures (Volney and Fleming 2007). As a result of these species-specific differences as well as the differences among forest types in which these insects are found and the uncertain (if present) interactions with weather, it seems quite unlikely that general relationships exist to connect insect outbreaks and fire risk. In light of these findings, we suggest that management should focus on developing and implementing models specific to the local system (i.e., insect species and forest type) and management objectives.

Uncertainties

We expected to find that the influence of cumulative defoliation on ignition would depend on fire weather. However, we did not identify any significant interaction effects between fire weather and defoliation at the scale of the ecoregions examined. This is likely because the context-dependent relationship between weather and defoliation was already well captured in our models using the GAM function that controls for temporal autocorrelation in fire activity. Additionally, we did not find a significant effect of lightning on the probability of ignition. This indicates simply that spatial variation in lightning strike frequency within each ecoregion $\times$ season combination did not influence our model selection procedure. Broader-scale models of ignition probability that include multiple ecoregions will likely still need to consider the role of lightning (Wotton and Martell 2005).

Forest structure is also expected to affect the relationship between defoliation and ignition. However, given the long time frame of our study it was not possible to include spatially explicit forest data at the same temporal scale. In dividing our analysis between two geographic regions, we endeavored to account for some expected variation in forest structure. Opportunities remain to include more detailed forest and physiographic information (e.g., fuel complexity, topography), both at the stand and landscape level, into models of fire ignition. Nonetheless, at the coarse scales examined our results indicate that cumulative defoliation has the potential to both increase and decrease the probability of ignition depending on the time scale and ecoregion examined.

Finally, it is important to note that we examined only one component of wildfire: ignition. Other empirical studies have examined the effects of insect activity on area burned (Hart et al. 2015), fire severity (Harvey et al. 2014, Meigs et al. 2016), fire occurrence (Lynch and Moorcroft 2008, Flower et al. 2014), and fire behavior (Simard et al. 2011). All of these elements of fire activity are relevant to understanding and managing the ecological and economic consequences of large wildfires. Interestingly, many of these studies have found that insect outbreaks reduce the risk of fire (e.g., Flower et al. 2014, Meigs et al. 2016) or do not affect it at all (e.g., Hart et al. 2015). Similarly, a spatial simulation study in the eastern spruce budworm system also found no evidence for an effect of defoliation on area burned (James et al. 2011). The lack of a general consensus regarding the effects of insect outbreaks on forest fires may be the result of the diversity of response variables and temporal scales over which those response variables have been analysed. For example, whereas we were able to identify a relationship between defoliation and the probability of ignition, we do not yet know if there is an effect on area burned. We suggest that researchers should be careful not to entirely dismiss the possibility of an interaction between insect outbreaks and fire activity due to the lack of an interaction between specific attributes of these disturbances at specific scales.

Within the eastern spruce budworm system, there is a great deal of opportunity to further develop our understanding of defoliation-fire dynamics. Recent work has shown that defoliation affects fuel structure and that fuels affect spread rates and crowning potential (Watt 2014). It is very likely that models that include historical cumulative defoliation may help us to better understand variation in these other processes. Further empirical and simulation work is currently underway to address these questions.

CONCLUSIONS

Many researchers have suggested that eastern spruce budworm outbreaks affect fire risk (Graham 1923, Swaine et al. 1924, Prebble 1950, Baskerville 1975, Stocks 1987, McCullough et al. 1998), although few have addressed this question empirically (Pech 1987, Stocks 1987, Fleming et al. 2002). Using historical defoliation, weather, and fire ignition data, we showed that defoliation by the spruce budworm can decrease the probability of ignition at short time scales (~1 yr) but increases this probability at intermediate time scales (8–10 yr). We also showed that the specific effect of defoliation on fire depends on local context including ecoregion and fire season. Given that climate change is expected to increase fire and insect activity in northern forests (Flannigan et al. 2009, Régnière et al. 2012), we expect that the inclusion of insect defoliation in fire risk models will become increasingly important for effective fire and forest management.

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